

Lab Manual — Question Solutions

Answers to the review / assignment / technical questions and worked numericals of all six experiments

Particulars	Details
Name of the student	
Enrolment No. · Branch	Robotics and Automation
Subject	Automated Manufacturing (BE05041021)
Experiments covered	All 6 (CAM & Automation · CNC · CAPP · AS/RS · CMM)
Date	

Experiment 1 — Introduction to Computer-Aided Manufacturing & Automation

Sample Problem — Control Resolution ($CR = p / n$)

STAR MILL axis: lead-screw pitch **p = 4 mm**, stepper motor **n = 200 steps/rev**.

$$CR = p / n = 4 \text{ mm} / 200 = 0.02 \text{ mm} = 20 \text{ micrometre}$$

The smallest commandable movement of the axis is 20 um per step.

Review Questions

1. What are the main limitations of traditional NC systems that led to the development of modern CNC machines?

Ans. Traditional NC used **hardwired control and punched tapes**. Its main limitations were: programs could not be edited at the machine (the tape had to be re-punched), the paper/mylar tapes wore out, tore and read unreliably, there was no program storage or memory, no diagnostics or error-checking, and repeated tape reading slowed production. **CNC overcame these** by using an onboard microcomputer (MCU) that stores and edits programs in memory, allowing easy editing, storage, diagnostics, interpolation and far higher reliability.

2. Explain the critical difference between machine accuracy and machine precision using a target diagram analogy.

Ans. **Accuracy** is how close the machine gets to the actual target (commanded) point — hitting the bull's-eye. **Precision (repeatability)** is how closely repeated moves cluster together, whether or not they hit the target. On a target: a tight group away from the centre is **precise but not accurate**; shots scattered around the centre are **accurate on average but not precise**; a tight group on the centre is **both**. A good CNC machine must be accurate and precise.

3. How does Intelligent Manufacturing adapt to production variations compared to standard flexible automation?

Ans. Standard **flexible automation** must be **reprogrammed by a human** to handle a new product or mix. **Intelligent Manufacturing** goes further: using sensors, real-time data, AI/machine learning and adaptive control, it **detects variations** (tool wear, material changes, dimensional drift) and **automatically adjusts the process** without human reprogramming. It is self-monitoring and self-optimising, so it adapts to production variation in real time instead of needing manual setup changes.

4. What role does a Direct Numerical Control (DNC) network play in an automated production facility?

Ans. In **DNC** a central computer is networked to many CNC/NC machines. It **stores and distributes part programs** to each machine on demand, **collects production and status data** back from them, and manages program version control and real-time data flow. This removes the need to load programs individually at each machine and **integrates the shop floor** into a computer-integrated manufacturing (CIM) system.

Experiment 2 — Motion Control, Machine Elements & Feedback Systems

Sample Problem — Linear Displacement ($x = N/E \times P$)

Rotary encoder **E = 2500 pulses/rev**, ball-screw pitch **P = 5 mm/rev**, pulses counted **N = 12 500**.

$$x = (N / E) \times P = (12500 / 2500) \times 5 = 5 \times 5 = 25 \text{ mm}$$

The table moves a linear distance of 25 mm.

Assignment Questions

1. How do recirculating ball screws eliminate the problem of "backlash" found in traditional Acme lead screws?

Ans. In a ball screw, hardened steel **balls recirculate between the screw shaft and the nut**, converting rotary to linear motion through **rolling contact**. Backlash (lost motion on reversal) is removed by **preloading** — a double nut or oversized balls keep the balls in firm contact with both flanks of the thread, eliminating the clearance/play that a sliding Acme lead screw has. The result is **near-zero backlash** and high positioning accuracy.

2. Compare the functional advantages of Linear Motion guideways versus Hardened and ground guideways in high-speed machining centres.

Ans. **LM (rolling) guideways** use recirculating rolling elements, giving very low friction, no stick-slip, high-speed capability and easy replacement — ideal for fast, precise positioning. **Hardened & ground (box) guideways** are sliding contact with a large surface area, giving high load capacity, stiffness and vibration/chatter damping, but with higher friction and lower speed. In high-speed machining centres **LM guides are preferred** for speed and precision, while box ways suit heavy roughing cuts.

3. Why might a high-precision CNC machine use Linear Glass Scales for positional feedback instead of relying solely on the servo motor's rotary encoder?

Ans. A rotary encoder on the motor/screw measures only the **screw's rotation**, so it cannot see errors introduced after the screw — ball-screw pitch error, thermal expansion,

backlash and wear. A **linear glass scale** mounted on the axis measures the **table's actual position directly** (true closed-loop feedback), so it detects and corrects all these errors, giving the higher positional accuracy that high-precision work demands.

4. What is the primary difference between Hydrostatic and Hydrodynamic bearings, and how does this affect spindle startup?

Ans. In a **hydrodynamic** bearing the oil film is generated by the **rotation of the journal itself**, so at startup (zero/low speed) there is metal-to-metal contact and wear until speed builds the film. In a **hydrostatic** bearing the oil is **pressurised by an external pump**, maintaining a supporting film even at zero speed. Therefore hydrostatic bearings give **zero-wear starts**, whereas hydrodynamic bearings wear slightly at every start and stop.

Experiment 3 — Manual Part Programming & Simulation (Turning & Milling)

Mathematical Drill — Spindle Speed from Cutting Speed

Cutting velocity $v = (\pi \cdot D \cdot N) / 1000$, rearranged for spindle speed N:

$$N = (1000 \times v) / (\pi \times D)$$

v = cutting speed (m/min), D = work/cutter diameter (mm), N = spindle speed (RPM)

Assignment Questions

1. Explain why tool nose radius compensation (G41/G42) is vital when turning tapered profiles or circular contours on the XL TURN lathe.

Ans. A turning tool is programmed at its **theoretical sharp point**, but the real cutting edge is a **rounded nose of finite radius**. On straight turning and facing this causes no error, but on **tapers and circular contours** the actual contact point shifts around the nose radius, producing profile errors (undercut or oversize). **G41/G42 nose-radius compensation** makes the control automatically offset the path by the nose radius, so the programmed profile is cut accurately.

2. What occurs on a CNC mill if a programmer accidentally swaps a G02 command for a G03 path when milling a corner arc?

Ans. **G02 is clockwise** and **G03 is counter-clockwise** circular interpolation. Swapping them makes the cutter travel the arc in the **wrong direction**, cutting a mirror-image/incorrect arc — it either **gouges into the part** or cuts on the wrong side, giving a wrong profile and a possible tool collision. The arc end-point may also become geometrically invalid for that direction, triggering a **radius/arc alarm**.

3. Differentiate between the physical coordinate referencing modes of G90 and G91 commands.

Ans. **G90 (absolute)** references every coordinate from a **fixed work origin (part zero)** — each value is the actual position to reach. **G91 (incremental)** references each coordinate from the **tool's current position** — each value is the distance and direction to move next. G90 is easy to check against the drawing; G91 is convenient for repeated or patterned moves.

4. What primary parameter determines whether a feed rate command uses feed-per-minute (G94) or feed-per-revolution (G95) parsing modes?

Ans. It is the **reference unit of the feed rate** — whether feed is tied to **time** or to **spindle rotation**. **G94 = feed per minute (mm/min)**, used mainly in milling. **G95 = feed per revolution (mm/rev)**, used mainly in turning, where the feed is linked to each revolution of the spindle.

Experiment 4 — Retrieval & Generative CAPP Systems

Problem Case — Spindle Speed from the Machinability Database

Database cutting speed **Vc = 150 m/min**, rough-turning a mild-steel rod of **D = 40 mm**:

$$N = (1000 \times Vc) / (\pi \times D) = (1000 \times 150) / (\pi \times 40) \\ = 150000 / 125.66 = \text{approx. } 1194 \text{ RPM}$$

Questions

1. Why is CAPP considered an indispensable link when transitioning a factory floor from basic islands of automation to a fully computer-integrated manufacturing (CIM) system?

Ans. **CAPP (Computer-Aided Process Planning)** is the **bridge between CAD (design) and CAM (manufacturing)**. Isolated "islands of automation" — standalone CAD, CNC, inspection, etc. — cannot share data. CAPP automatically converts design data into **standardised, consistent process plans and routing sheets** and feeds them to scheduling, CNC and material handling. Without it there is a manual, inconsistent gap between design and the shop floor, so CAPP is essential to **tie everything together into a full CIM system**.

2. Under what specific manufacturing volume and variety scenarios would a company choose a Retrieval (Variant) CAPP system over a Generative CAPP system?

Ans. **Retrieval (Variant) CAPP** suits **high volume with low variety**, where parts fall into established **part families** that already have master process plans — the existing plan is simply retrieved and lightly edited, which is fast and low-cost. **Generative CAPP** suits **high variety / low volume** or new, unique parts where no master plan exists and a plan must be built from scratch. So a company making **many similar parts with stable designs** chooses the Variant system.

Experiment 5 — Material Handling, AS/RS & Data Capture

Analytical Drill — AGV Cycle Time & AS/RS Capacity

Part A — AGV cycle time: $v_c = 60 \text{ m/min}$, $L_d = 120 \text{ m}$, $L_e = 90 \text{ m}$, $T_l = 0.8 \text{ min}$, $T_u = 0.7 \text{ min}$.

$$T_c = T_l + T_u + L_d/v_c + L_e/v_c \\ = 0.8 + 0.7 + 120/60 + 90/60 \\ = 0.8 + 0.7 + 2.0 + 1.5 = 5.0 \text{ minutes}$$

Part B — AS/RS storage capacity: Total Storage Capacity = $2 \cdot n_a \cdot x \cdot y$ (where n_a = number of aisles, x = horizontal compartments, y = vertical tiers; the factor 2 accounts for a rack on **both sides** of every aisle).

Assignment Questions

1. Which Principle of Material Handling focuses specifically on reducing dead-weight and maximising space utilisation in a facility?

Ans. The **Space Utilisation Principle**. It states that all available space must be used effectively — making the best use of the building's **cubic volume** through high-density and vertical storage (e.g. tall AS/RS racks) and minimising wasted floor space and dead volume. (The related **Unit-Load Principle** addresses keeping the dead-weight of handling units low relative to the payload moved.)

2. Explain how a vehicle collision-avoidance loop works when managing a fleet of multiple AGVs on a shared track layout.

Ans. The guide path is divided into **zones or blocks**, and a traffic controller allows **only one AGV in a block at a time**. An AGV must **request and be granted permission** before entering the next block; if the block ahead is occupied it **waits**. Combined with onboard proximity/bumper sensors and defined right-of-way rules at intersections, this **blocking-zone logic** prevents two vehicles from occupying the same path segment, avoiding collisions and deadlock.

3. What are the primary performance advantages of implementing an AS/RS compared to a conventional manual forklift-and-pallet rack system?

Ans. An AS/RS offers **much higher storage density** (tall, narrow aisles use cubic space far better), **faster automated 24/7 storage and retrieval** with higher throughput, **greater accuracy and real-time inventory control** (fewer lost or mis-picked items), **reduced labour and labour cost**, and **improved safety** (fewer human-forklift accidents). A manual forklift system is slower, needs wide aisles, is labour-intensive and more error-prone.

4. Why are 2D matrix codes (such as QR codes) increasingly favoured over legacy 1D linear barcodes in harsh automated manufacturing environments?

Ans. 2D matrix codes store **much more data in a smaller area**, can be read from **any orientation (omnidirectional)**, and most importantly have **built-in error correction (redundancy)**, so they can still be decoded when partly damaged, dirty, scratched or poorly printed. In harsh conditions (dust, oil, abrasion) a 1D barcode fails once its bars are smudged or damaged, whereas a **2D code stays readable**, making it far more robust.

Experiment 6 — Dimensional Inspection & Calibration using CMM

Sample Problem — Mean Positioning Error of a CMM

A 50.000 mm gauge block is measured four times: 50.002, 50.005, 50.003, 50.002 mm.

$$\begin{aligned} \text{Mean } \mu &= (50.002 + 50.005 + 50.003 + 50.002) / 4 \\ &= 200.012 / 4 = 50.003 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Mean positioning error } E_{\text{pos}} &= | \mu - X_{\text{nom}} | \\ &= | 50.003 - 50.000 | = 0.003 \text{ mm} = 3 \text{ micrometre} \end{aligned}$$

$$\text{Repeatability } \sigma = \sqrt{ \sum (x_i - \mu)^2 / (n - 1) } = \text{approx. } 1.41 \text{ micrometre}$$

Technical Questions

1. Compare contact CMM inspection methods with non-contact laser scanning in terms of measuring speed, point-cloud density, and accuracy limits.

Ans. A **contact (touch-trigger) CMM** takes **discrete points one at a time**, so it is **slower with low point density**, but each point is **highly accurate** (micron level) — best for precise features such as bores and distances. **Non-contact laser scanning** projects a laser line/point cloud and captures **thousands of points per second**, giving **very high speed and very high point-cloud density** ideal for complex 3D surfaces and reverse engineering, but with **lower accuracy** (tens of microns) and sensitivity to surface finish and reflectivity.

2. Explain how environmental conditions (such as ambient room-temperature variations) introduce measurement errors in high-precision coordinate metrology.

Ans. Metals **expand and contract with temperature**. If the room differs from the **20 °C reference standard**, both the workpiece and the machine's scales and structure change dimension, so the measured values **drift from the true values**. Temperature **gradients** also distort the CMM frame unevenly. Because even a 1 °C change causes micron-level errors over typical part sizes, precision metrology is performed in **temperature-controlled rooms (20 ± 1 °C)** or with active temperature compensation.